

Nakorn Boon Han Charayaphan

LEAD ENGINEER · PHYSICAL AI

 Singapore

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Job Experience

Lead Engineer

Singapore

HOME TEAM SCIENCE AND TECHNOLOGY AGENCY (HTX)

June 2025 - Present

- Team Lead to develop Agentic and World Modelling capabilities for Physical AI
- Development Lead for Physical AI Robot Skills Development, in collaboration with Mistral AI
- Led Cross Team Collaboration efforts to build PoC and showcases for HTX events
- Deployed Compute and Common Services infrastructure (SLURM, Git, Container Registries) for development

Platform Engineer

Singapore

INTERNATIONAL BUSINESS MACHINES

March 2022 - June 2025

- Deployed and Configured AWS Cloud and Bare Metal RHEL in airgapped environments, hosting ML Workloads for End Users
- Developed Security and Compliance OpenShift and Cloud Deployments, clearing them for Production
- Developed IaC and CI/CD pipelines on AWS and GitLab, making infrastructure and code repeatable and auditable

DevOps Engineer

Singapore

OPEN SOURCE ROBOTICS CORPORATION

March 2021 - March 2022

- Development and Deployment Engineering of Robotics Integration Software
- Operationalized in Changi Airport and Changi General Hospital

Robotics Engineer

Singapore

CENTER FOR HEALTHCARE ASSISTIVE AND ROBOTICS TECHNOLOGY

June 2019 - March 2021

- Robotics Test Lab in healthcare domain, building mobile robot prototypes and systems testing for Multi Robot Systems

Projects

Dreamworld

 cnboonhan/dreamworld_gs

AGENTIC HARNESS AND 3D WORLD MODELLING PROOF OF CONCEPT

2026

- Agents demonstrate solving long horizon physical tasks through Harness Engineering
- 3D World Modelling replaces site access and hardware, cutting the cost of testing logistics

Humanoid Skill Library

END TO END PHYSICAL AI HUMANOID ROBOT SKILLS DEVELOPMENT

2026

- Building Robot Skills, such as Door Opening and Lift Taking, learned from demonstration data
- More scalable by collecting more data, instead of building more complex robot systems
- Needs no changes to the environment: the Robot interacts with buildings as Humans do

Systems Experience

Cloud & Platform

AWS
OpenShift & Red Hat Enterprise Linux
GitLab
Keycloak OIDC

Physical AI

Multimodal Agentic Models
3D Simulation and World Modelling
Vision Language (Action) Models
Humanoid Robot Data Collection

Education

Masters of Computing	Singapore
NATIONAL UNIVERSITY OF SINGAPORE, GPA 4.22/5.00	2024
Bachelors of Engineering, Artificial Intelligence and Software Engineering	United Kingdom
UNIVERSITY OF EDINBURGH, FIRST CLASS HONOURS	2019

Certifications

Red Hat Certified Engineer	2024
Certified Kubernetes Administrator	2024
AWS Certified Machine Learning	2024
AWS Certified DevOps Engineer - Professional	2023